

Inhibitors of nucleic acid synthesis (Quinolones and folate antagonists)

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Objectives

- Compare and contrast nucleic acid inhibitors: mechanisms of action, indications and their adverse effects.
- Formulate appropriate nucleic acid inhibitors used for different infections with special emphasis on their contraindications and side effects.

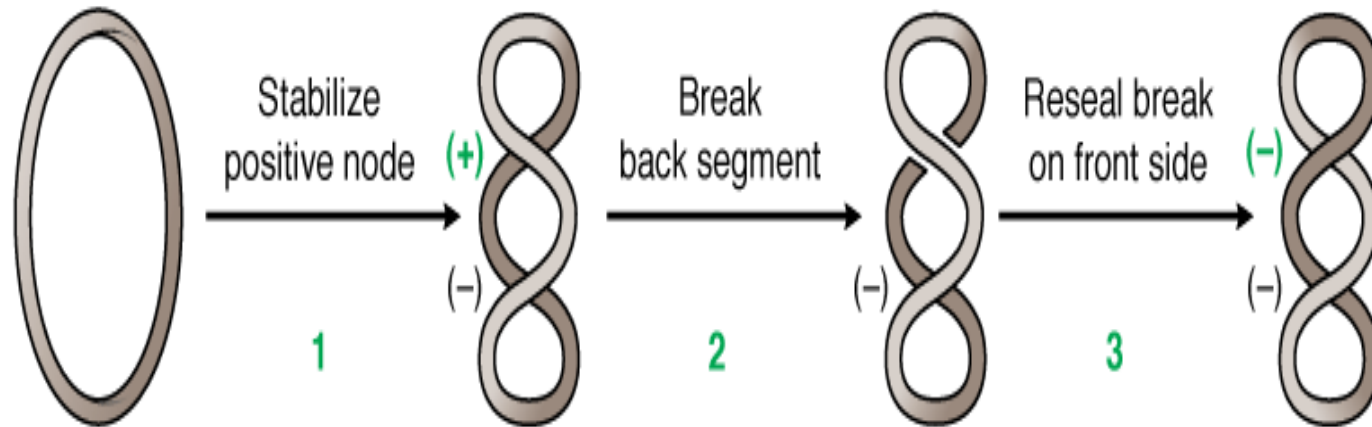
Inhibitors of nucleic acid synthesis

- I-Quinolones

- **1st generation:** Nalidixic acid
- **2nd generation:** Fluoroquinolones:
 - -Ciprofloxacin
 - -Ofloxacin
 - Norfloxacin
 - -Pefloxacin
- **3rd generation:**
 - -**Levofloxacin**(tavanic)
 - Gatifloxacin
 - - Sparfloxacin
- **4th generation:**
 - Trovafloxacin , **Moxifloxacin** (avelox)

Mechanism of action:

- -Bactericidal.
- -Inhibit [DNA gyrase \(topoisomerase II\)](#) which is responsible for the unwinding of double stranded DNA leading to inhibition of DNA replication (more significant in gram **-ve** organism). Also inhibit [topoisomerase IV](#) that is required for cell division (significant in gram **+ve** organisms).



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*:
www.accessmedicine.com

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Model of the formation of negative DNA supercoils by DNA gyrase. The enzyme binds to two segments of DNA (1), creating a node of positive (+) superhelix. The enzyme then introduces a double-strand break in the DNA and passes the front segment through the break (2). The break is then resealed (3), creating a negative (-) supercoil. Quinolones inhibit the nicking and closing activity of the gyrase and, at higher concentrations, block the decatenating activity of topoisomerase IV. From Cozzarelli NR. DNA gyrase and the supercoiling of DNA. *Science*, 1980, 207:953-960. Reprinted with permission from AAAS.

Antibacterial spectrum and uses

- -Mainly against ***gm –ve organisms***, e.g. pseudomonas, H. influenza, E. coli and proteus, Moraxella, Chlamydia, Legionella (accumulate in macrophages and PMN).
- -Effective against penicillinase producing ***gonococci***, less effective against gm +ve organisms (staph, pneumococci).

Cont.,

- **-According to generations:** 2nd generation exhibits activity against gm –ve aerobes, some +ve organisms & atypical mycoplasma, while 3rd generation shows increased activity against gm +ve and atypical bacteria and some anaerobes. 4th generation is active against **anaerobic** as well as gm +ve organisms(& maintain gram –ve activity).

Uses:

- Typhoid and traveler's diarrhea.
- **Urinary tract** infections (gm –ve bacilli).
- **Prostatitis** (norfloxacin and levofloxacin).
- **Gonorrhea** *not* **syphilis** (ofloxacin; single dose), also levofloxacin.
- **Respiratory** tract infection (**S. pneumonia**) e.g. levofloxacin, trovafloxacin, moxifloxacin and gatifloxacin (so-called *respiratory fluoroquinolones*) *not* ciprofloxacin.
- **Bone** and soft tissue infection.

Ciprofloxacin :

- Is the drug of choice for the prophylaxis and treatment of **anthrax**.
- Also used as *alternative to aminoglycosides* or *synergistically* with β -lactam (e.g. ampicillin).
- The **most potent** of all quinolones for **Pseudomonas infections**.
- Effective against *traveler's diarrhea and in TB*.
- **Effective against systemic infections, Not** against MRSA, enterococci and pneumococci.



- Norfloxacin
- is effective against gram –ve and +ve organisms (*urinary tract infections and prostatitis*), not against *systemic infections (insufficient plasma levels)*.
- Cross- resistance (altered target or decreased accumulation) exist among quinolones.



- Levofloxacin:
- Has broad spectrum of activity e.g. skin infections, acute sinusitis and chronic bronchitis.
- Used in treatment of **prostatitis** and sexually transmitted diseases (not syphilis).
- Excellent in treatment of respiratory infections due to ***S. pneumonia***.
- Moxifloxacin:
- active against **anaerobic** as well as gram +ve organisms but ***very low activity against P. aeruginosa***.

Antimicrobial spectrum of quinolones

First Generation

Gram (+) cocci
Gram (+) bacilli
Gram (-) cocci
Gram (-) rods
Anaerobic organisms
Spirochetes
Mycoplasma
Chlamydia
Other

First-generation quinolones, which are used less often today, have moderate gram-negative activity. They achieve minimal serum concentrations and are restricted to the treatment of uncomplicated urinary tract infections.

Third Generation

Gram (+) cocci
Gram (+) bacilli
Gram (-) cocci
Gram (-) rods
Anaerobic organisms
Spirochetes
Mycoplasma
Chlamydia
Other

Third-generation fluoroquinolones retain expanded gram-negative activity and show improved activity against atypical organisms and specific gram-positive bacteria.

Second Generation

Gram (+) cocci
Gram (+) bacilli
Gram (-) cocci
Gram (-) rods
Anaerobic organisms
Spirochetes
Mycoplasma
Chlamydia
Other

Second-generation fluoroquinolones have expanded gram-negative activity and also have some activity against gram-positive and atypical organisms, such as Mycoplasma pneumoniae and Chlamydia pneumoniae.

Fourth Generation

Gram (+) cocci
Gram (+) bacilli
Gram (-) cocci
Gram (-) rods
Anaerobic organisms
Spirochetes
Mycoplasma
Chlamydia
Other

Fourth-generation fluoroquinolones shows improved gram-positive coverage, maintains gram-negative activity, and gains anaerobic coverage.

Typical therapeutic applications of fluoroquinolones

ANTHRAX

- *Ciprofloxacin* is the drug of choice for postexposure prophylaxis and for treatment of anthrax. Alternate agents include *doxycycline*.

Gram (+) cocci

Streptococcus pneumoniae

Gram (+) bacilli

Bacillus anthracis

Gram (-) cocci

Gram (-) rods

Enterobacter species

Escherichia coli

Haemophilus influenzae

Klebsiella pneumoniae

Legionella pneumophila

Proteus mirabilis

Pseudomonas aeruginosa

Serratia marcescens

Shigella species

Anaerobic organisms

Spirochetes

Mycoplasma

Chlamydia

Other

URINARY TRACT INFECTIONS

- *Ciprofloxacin* and *levofloxacin* are effective in treating uncomplicated and complicated urinary tract infections.

RESISTANT RESPIRATORY INFECTIONS

- *Levofloxacin* is often effective in treating infections unresponsive to β -lactam antibiotics, such as *amoxicillin-clavulanic acid*.
- *Ciprofloxacin* is not the drug of first choice for pneumonia or sinusitis, because it has weak activity against *Streptococcus pneumoniae*, a common causative agent.
- *Levofloxacin* and *moxifloxacin* are known as "respiratory fluoroquinolones" due to their activity versus *Streptococcus pneumoniae*

GASTROINTESTINAL TRACT INFECTIONS

- *Ciprofloxacin* is highly efficacious in treating acute diarrheal illnesses due to enteric pathogens.

- **Pharmacokinetics:**

- *Not completely absorbed when orally administered (ranging from 35-70% for norfloxacin to 85-95% for others).*
- Also Parenterally. Binding to pp is 10-40 %.
- ***Levofloxacin*** and ***moxifloxacin*** have the *longest* half lives which permit ***once daily*** dosing.

Cont.,

- Absorption of quinolones is **affected by** sucralfate, antacids- containing aluminum, magnesium or dietary supplement containing iron, zinc and calcium. Well distributed into all tissues.
- Penetration to CSF is low **except** for **ofloxacin**.
- Excretion is via renal route(dose adjustment in renal dysfunction). **Moxifloxacin** is excreted mainly by **liver** so no dose adjustment in renal dysfunction.

Adverse effects:

- GIT disturbances: the most common.
- CNS symptoms, e.g. headache, dizziness. Should be used cautiously in patients with CNS disorders.
- Ciprofloxacin interferes with absorption of theophylline and may evoke seizures.
- **Phototoxicity** (should interrupt therapy).



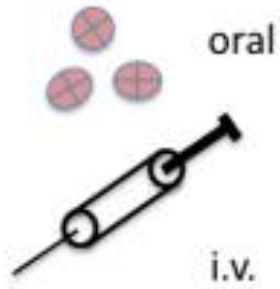
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- **Nephrotoxicity & Liver** toxicity with **trovafloxacin** (reserved for serious infections, not more than 14 days and only in hospital).

- Arthropathy (cartilage damage and tendinitis) (in children less than 18 years), more over 60 years, steroid therapy, heart and kidney and lung transplantation.

Fluoroquinolone Uses

Routes:



Ophthalmic infections

*Respiratory Infections,
Inhaled Anthrax,
Mycobacteria (TB)*

*Infections of bone,
joints & soft tissues*

*GI & Abdominal
Infections*

*Prostatitis
UTIs (MDR strains)
& STDs (Chlamydia)*

Mechanism:

Inhibits DNA Gyrase & Topoisomerase IV



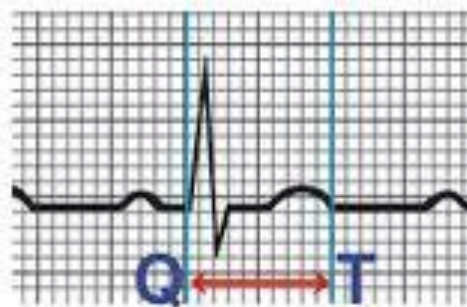
bactericidal

Adverse Effects:

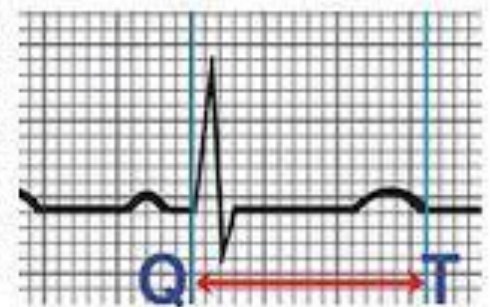
- Tendon rupture
- Children <18 yo (cartilage)
- Pregnancy Category C
- Seizures, prolong QT
- Dizziness, Confusion
- Photosensitivity

Contraindications

- **Contraindicated** *in pregnancy, lactation and patients less than 18 years* (may lead to arthropathy).
- **Sparfoxacin**, gatifloxacin, levofloxacin, and moxifloxacin may ***prolong QT interval*** and should not be used in patients predisposed to dysrhythmias.



Normal Q-T interval



Long Q-T interval

Drug interactions

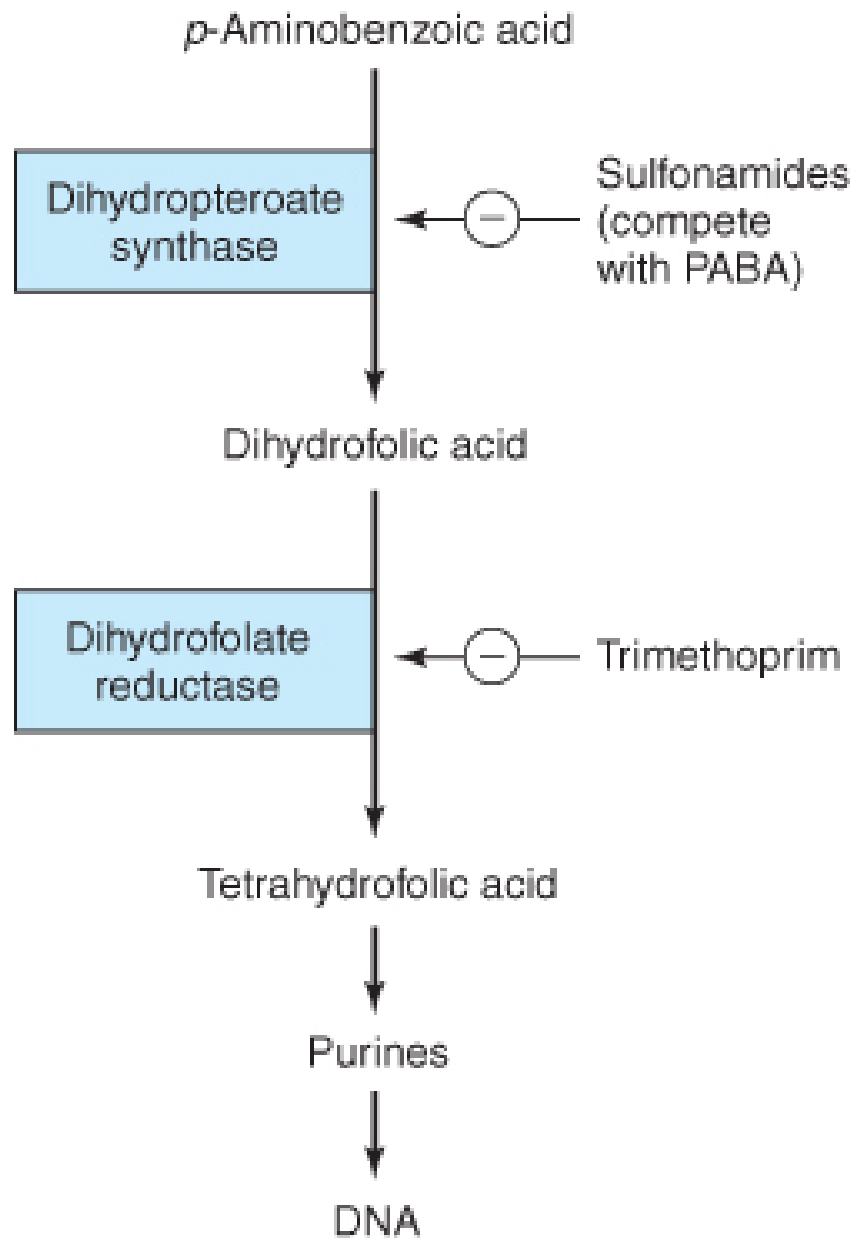
- **Absorption** is affected by antacids and cations.
- Cimetidine interferes with elimination of quinolones.
- Metabolism of theophylline is inhibited by ciprofloxacin, while 3rd and 4th generations increase concentrations of warfarin, caffeine and cyclosporine.

Resistance

- **Altered target:** Mutations in the bacterial DNA gyrase have been associated with a decreased affinity for fluoroquinolones. Resistance is frequently associated with mutations in *both DNA gyrase and topoisomerase IV*.
- **2. Decreased accumulation:** Reduced intracellular concentration of the drugs .
- One involves a *decreased number of porin proteins in the outer membrane* of the resistant cell, thereby impairing access of the drugs to the intracellular topoisomerases.

Folate antagonists

- I-Sulfonamides:
- Members: Sulfadiazine, sulfamethoxazole, sulfadoxine, sulfacetamide (topical eye preparation), sulfisoxazole, and sulfasalazine.
- -Mechanism of action:
- -Bacteriostatic.
- -Structural analogues of **PABA**. So they compete with it for the enzyme dihydropteroate synthase, leading to inhibition of folate synthesis with inhibition of DNA and RNA synthesis.
- N.B.: human cells utilize already formed folic acid; this explains the selectivity of sulfonamides to bacteria.



- Pharmacokinetics:
- -All are orally absorbed except **sulfasalazine** (that is why it is used in chronic inflammatory bowel diseases).
- -**High pp bound** (depends on pKa), cross BBB and placental barrier.
- -**Acetylated** in liver (slow acetylators are liable to toxicity of sulfapyridine) into metabolites which causes **crystalluria** (*at neutral and acidic pH*).
- -Excreted by glomerular filtration (affected by renal dysfunction).

Therapeutic uses:

- **Eye** infections: topical **sulfacetamide**.
- **Burns**: topical silver **sulfadiazine** (sulfamafenide causes pain and acid –base disturbances).
- **Ulcerative colitis**: **sulfasalazine**(sulfasalazine is a prodrug composed of **5-aminosalicylic acid (5-ASA)** linked to **sulfapyridine** (not absorbed).
- **Toxoplasmosis** :(**sulfadiazine**)
- **chloroquine-resistant malaria**: **sulfadoxine** in combination with **pyrimethamine**.
- **Urinary tract infection** (***enterobacteria*** ***and nocardia***).Trimethoprim or co-trimoxazole are preferred.

- Adverse effects:

- **Crystalluria and nephropathy:** due to insoluble metabolite precipitation. Can be avoided by increased **fluid** intake, **alkalinization** of urine and use of more **soluble** agents as sulfisoxazole and sulfamethoxazole.



- **-Hypersensitivity** reactions:
 - Skin rash, urticaria, erythema Multiform -severe (**Stevens-Johnson syndrome**)
- **-Hemopoietic disturbances:** Granulocytopenia and thrombocytopenia. Hemolytic anemia in patients with **G-6-P D.**

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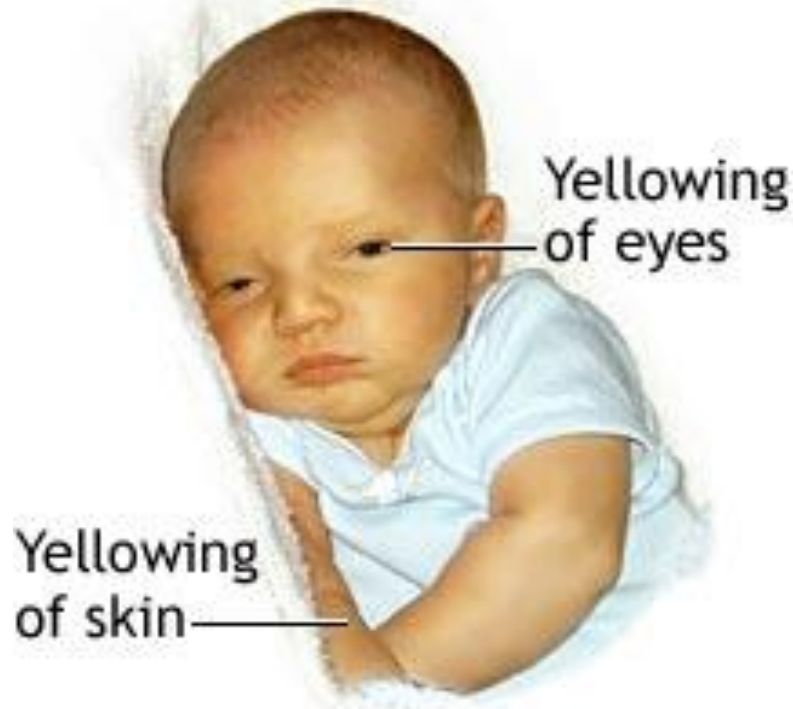
- **Kernicterus**: sulfonamides displace bilirubin from plasma protein. Free bilirubin crosses BBB (immature in newly born) and reaches CNS, thus contraindicated in infants less than 2 months and in pregnant women at term.

-Drug interactions:

- Increased plasma level of oral hypoglycemic drugs and anticoagulants due to plasma protein *displacement* (↑ level of methotrexate).

Kernicterus

Jaundice



Kernicterus



- Trimethoprim
- Mechanism of action:
 - - 20-50 folds more potent than sulfonamides.
 - -Inhibition of **dihydrofolate reductase enzyme** which converts folic acid to tetrahydro-folic acid (folinic acid) which is essential for DNA synthesis.
 - -Trimethoprim is about 50,000 times less efficient in inhibition of mammalian dihydrofolic acid reductase which explains drug **selective toxicity**.
 - -Antibacterial spectrum is the same as sulfamethoxazole.

Cont.,

- -Used in UTIs, prostatitis and vaginitis.
- Adverse effects:
- -**Megaloblastic anemia** due to folate deficiency (avoided by folic acid administration), granulocytopenia & leucopenia (especially *in pregnant women* and patients with very poor diet).
- This can be reversed by simultaneous administration of ***folic acid***.

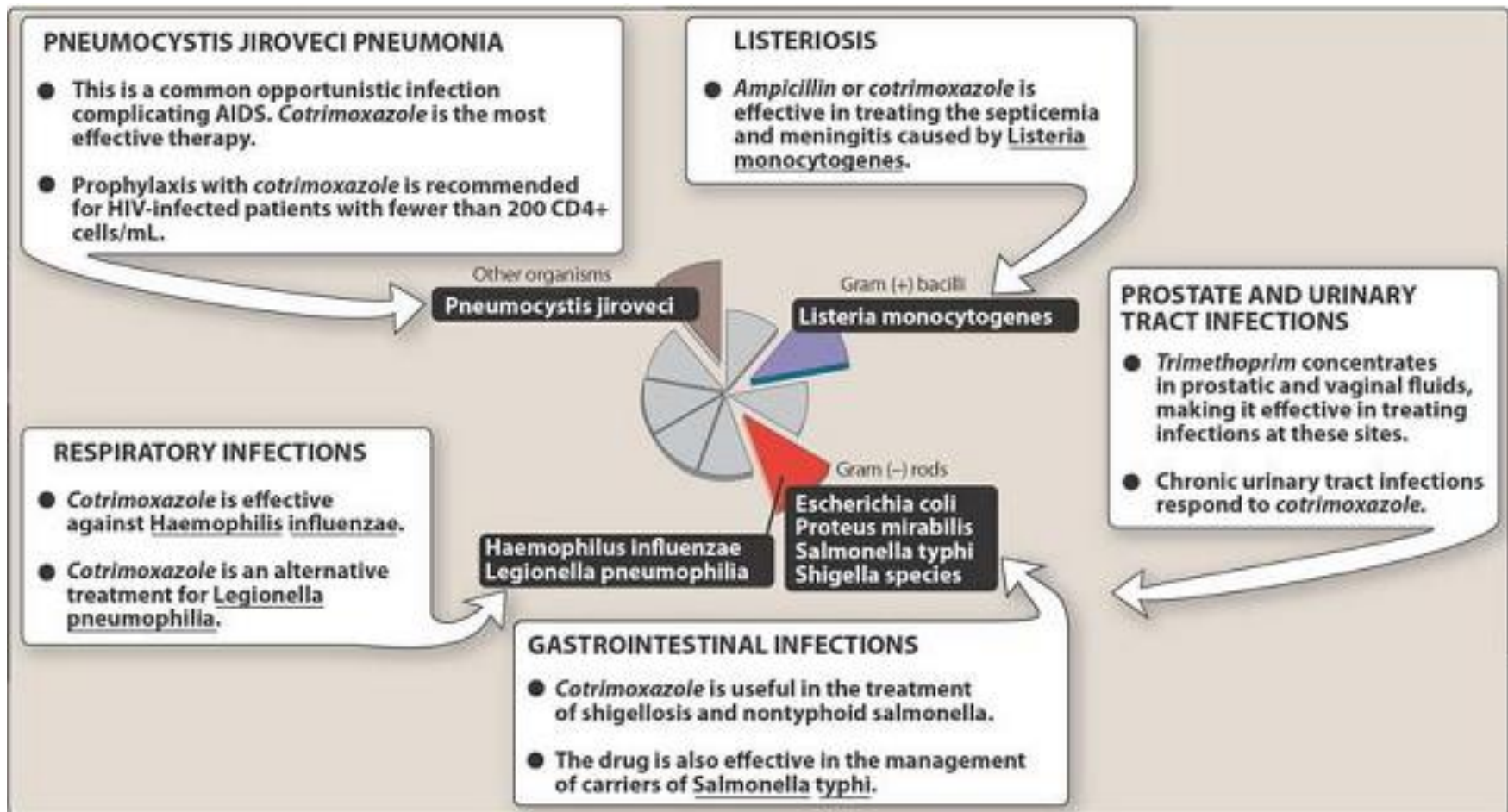
- **III- Co-trimoxazole:**
- It is a combination of sulfamethoxazole (400 mg) + trimethoprim (80 mg) 5:1 .Bacteriostatic.
- **Mechanism of action:**
- This **synergistic** combination is due to inhibition of two subsequent steps in the synthesis of folic acid.
- **Advantages:**
- More **potent**
- **Synergistic** combination.
- Less and delayed bacterial **resistance**.
- **Wider** spectrum against urinary and respiratory infections (Pneumocystis pneumonia, proteus, salmonella, shigella, H. influenza and gonococcus).



Therapeutic uses:

- **Urinary tract** infections, gonococcal urethritis, prostatitis and vaginitis.
- **GIT** infections: Salmonella and shigella.
- **Respiratory tract** infection due to H. influenza and S. pneumonia.
- Pneumocystis jirovecii pneumonia in AIDS:
the most effective
– therapy (IVI).





Typical therapeutic applications of co-trimoxazole

- Adverse effects:
- -(Same as sulfonamides and trimethoprim).
- -Dermatologic: very common and may be serious in elderly.
- -Hematological:
 - ***Megaloblastic anemia*** due to folate deficiency (avoided by folinic acid administration),
 - Granulocytopenia and leucopenia.
 - ***Hemolytic anemia*** in G-6-phosphate deficient patients.
- GIT disturbances.
- In AIDS: fever, rashes, diarrhea and /or pancytopenia.
- Interactions with phenytoin, methotrexate, and warfarin